

Baseline Transcriptomic Profiles of Healthy Liver and Small Intestine Tissue

Identify Tissue-Specific Immune and Coagulation Pathways Relevant to Microplastic Exposure Response.

ABSTRACT

Microplastics are ubiquitous contaminants, varying greatly in size and material, of both terrestrial and marine environments. They are becoming more frequently detected in both wildlife and humans. How they impact the health of living organisms is a major health and ecological concern. It is not yet fully clear how different tissues are prepared to respond to microplastic exposure, but an important place to look are the primary sites of exposure: the small intestine (site of absorption) and the liver (detoxification). This study analysed RNASeq data from healthy liver and small intestine in order to establish a baseline transcriptomic profile that may suggest how these tissues may respond to exposure. The results show that there are clear differences in gene expression between the two tissues which reflect the functions and gene pathways that may respond to microplastic exposure. The baseline transcriptome reported here may provide a more complete picture of the preparedness of the tissues to defend against microplastic exposure.

INTRODUCTION

Microplastics are increasingly being detected in human tissues, which raises a major public health concern (Malik, et al., 2025; Ragusa et al., 2022; Horvatus et al., 2022). They have been shown to trigger oxidative stress, inflammation and alteration of signaling pathways in human cell lines (Bahudar et al., 2023) as well as alteration of blood clotting indicators (activated partial prothrombin time and C-reactive protein) and the clotting factor, fibrinogen (Lee et al., 2024).

Microplastics are fragments of plastic debris that are less than 5 mm in size (Thompson et al., 2024). What they are composed of varies, and includes particles derived from the breakdown of larger plastic containers and materials; and microplastics used in cosmetics, textiles, and paints (Diana et al., 2025; Thompson et al., 2024; Wang et al., 2020). These fragments get dispersed by wind and water and can enter the biosphere when consumed or inhaled by organisms (Yang and Wang, 2023, Allen et al., 2019).

The primary site of absorption of ingested microplastic particles is the small intestine, while the liver is the major organ involved in detoxification of these particles (Stock et al., 2021, Thompson et al., 2024). There is limited research on how these tissues are prepared to handle microplastic exposure, although there are functions and proteins in each of these tissues that are involved in absorption, physical barrier, immunity/inflammation, and detoxification, which would likely be involved in handling microplastic exposure.

Small intestine tissue is primarily composed of four main specialized cell types: enterocytes, which are columnar epithelial cells; goblet cells, which secrete mucous; paneth cells, which secrete antimicrobial peptides; and enteroendocrine cells, which secrete hormones (Peterson and Artis, 2014). Some important proteins that may be important players in small

intestinal response to microplastics include DEFA5 and DEFA6, for example, which are defensin proteins, thought to be involved in responding to pathogen infections (Bevins and Salzman, 2011); and MUC2, a mucin protein, has been shown to be disrupted in the mucosal barrier in the presence of microplastics (Niu et al., 2025).

The liver is the primary site of processing and detoxification of absorbed substances from the small intestine (Trefts et al., 2017; Zanger and Schwab, 2013, Manikandan, P., & Nagini, S., 2018). Several proteins are synthesised in the liver, such as albumin (ALB), which is synthesised by hepatocytes and helps transport molecules in the blood, including nanoparticles for drug delivery (Qu et al., 2024); and Cytochrome P450 family members, which are involved in xenobiotic metabolism (Manikandan et al., 2018).

This differential gene expression analysis of liver and small intestine is aimed at examining healthy tissue transcriptomic profiles in order to identify baseline expression of genes involved in microplastic metabolism, oxidative stress and inflammation. This data may help to make predictions about how these tissues will respond to microplastics. Baseline gene expression will differ between small intestine and liver, with the liver showing higher expression of genes involved in metabolism and processing of xenobiotic particles, while the small intestine will show higher expression of genes involved in digestion, immune defense and formation of a mucous barrier.

METHODS

Data Acquisition and Processing.

RNA-seq data for healthy human liver and small intestine samples were obtained from the Delaware Biotechnology Institute's Biomix High Performance Computational Cluster. The dataset included three liver samples (liver_a, liver_c, liver_d) and 4 small intestine samples (si_4a, si_4b, si_4c, and si_4d).

Raw sequencing reads were assessed for quality using FastQC (version not recorded) (Andrews, 2010). Adapter sequences and bases of low-quality were removed using Trim Galore (Kreuger, 2015), which uses Cutadapt version 4.4 (Martin, 2010), which trimmed reads with a phred quality score <20. Trimmed reads were aligned to the human reference genome (GRCh38) using STAR aligner version 2.7.11b (Dobin et al., 2013). The resulting BAM files were sorted and used for downstream analysis in RStudio version 4.5.3 (Posit, 2026). Quality of the alignment was evaluated based on mapping rates which was ~92%, showing that the sequencing data was of high quality. Gene-level counts were obtained using HTSeq-count (Anders, et al., 2015) need version!).

Data Analysis

Downstream analyses were performed in RStudio (Version 2026.01.1+403 (2026.01.1+403)). Differential expression analysis was performed using the Bioconductor package edgeR (Robinson et al., 2010) version 4.8.2). A DGEList object was created to store gene IDs as row names. Genes with very low expression were filtered out first using the filterByExpr() function which removed genes that did not have a sufficiently large enough count for the analysis (tissue group was specified as the grouping variable).

Multidimensional scaling (MDS) analysis was performed using `plotMDS()` to visualize relationships between samples and gene expression patterns were visualized by calculating log-CPM values with `cpm(y, log=TRUE, prior.count=2)`.

Dispersion was estimated with the `estimateDisp()` function in edge R after using an exact test with `exactTest()` function. A generalized linear model (GLM) was also done and a design matrix was generated using tissue groups. Biological variability was determined using a biological coefficient of variation (BCV) plot (`plotBCV()`). The genes were considered to be differentially expressed based on adjusted p-values (FDR) with a threshold of <0.05 using `decideTests` and `topTags` in edgeR.

Data Visualization.

A heatmap was generated to visualize the 50 most variable genes, based on variance across the samples and using pheatmap package (Kolde, 2019) version 1.0.13). Smear plots were generated for exact test and GLM results using `plotSmear()` in order to visualize the relationship between log fold change and gene expression. The differentially expressed genes were highlighted in red.

All analyses were performed in R (R Core Team, 2025) version 4.5.3) using RStudio (Version 2026.01.1+403 (2026.01.1+403) on a MacOS Sequoia 15.7.4. The packages used were: edgeR (version 4.8.2); limma (Ritchie et al., 2015) version 3.66.0); statmod (version 1.5.1); and pheatmap (1.0.13).

Functional Enrichment Analysis

Functional enrichment analysis was performed to identify biological pathways and processes associated with the tissue-specific gene expression. Genes were considered significantly

differentially expressed if the FDR was less than 0.05. These genes were grouped by upregulation in liver tissue (negative logFC) or upregulated in small intestine (positive logFC). The lists of genes were separately entered into Enrichr (Chen et al., 2013; Kuleshov et al., 2016) for analysis and Gene Ontology Biological Processes and KEGG pathway terms were identified. Enrichment significance was based on adjusted p-values.

RESULTS

RNA-seq data from three healthy human liver tissue samples and 4 healthy human small intestine samples were analyzed to establish baseline transcriptome differences between tissues. Raw reads were checked for quality, adapter-trimmed and then aligned to the human reference genome (GRCh38). They were quantified at the gene level before analysing for differential expression with EdgeR.

Low-expression genes were filtered out and then transcriptomic relationships between the two tissue types were assessed using multidimensional scaling. Differential expression was evaluated by exact tests and generalized linear modeling. Once differentially expressed genes were identified, functional enrichment analysis was performed using Enrichr to identify biological pathways associated with tissue-specific transcription.

Table 1. Sequencing quality control, trimming and alignment summary for liver and small intestine RNA-seq samples. The percentage of bases trimmed and uniquely mapped reads are shown for each sample. The high read retention and high mapping efficiency (88.19-93.76% alignment to reference genome) indicates that the quality of the sequences were strong.

Sample	Raw Reads	Reads Retained	Bases Trimmed (%)	% Uniquely Mapped
liver_a	20,000,000	20,000,000	4.4	92.26
liver_c	12,225,037	12,225,037	4.0	93.76
liver_d	20,000,000	20,000,000	10.7	92.91
smallintestine_4a	18,593,490	18,593,490	2.0	88.47
smallintestine_4b	20,000,000	20,000,000	1.9	89.65
smallintestine_4c	20,000,000	20,000,000	2.1	88.19
smallintestine_4d	19,505,044	19,505,044	1.9	90.08

Global Expression Patterns

Multidimensional scaling (MDS) analysis showed a clear separation between liver and small intestine samples, indicating a clear difference in gene expression. Samples clustered with their tissue type groups along the Leading logFC dim 1 (87%) axis. The liver samples showed the most consistency in expression profiles, while the small intestine samples did exhibit some variability along the Leading logFC dim 2 (4%) axis (Fig. 1). However, the scale on the dim1 axis was smaller than the dim 1 axis (-2.0 to 1.0 vs -6 to 4). The biological coefficient of variation (BCV) plot shows a decrease in variability as gene expression increases. Low count

genes had more variability, while higher count genes had less variability, as expected in RNASeq data (Fig. 2).

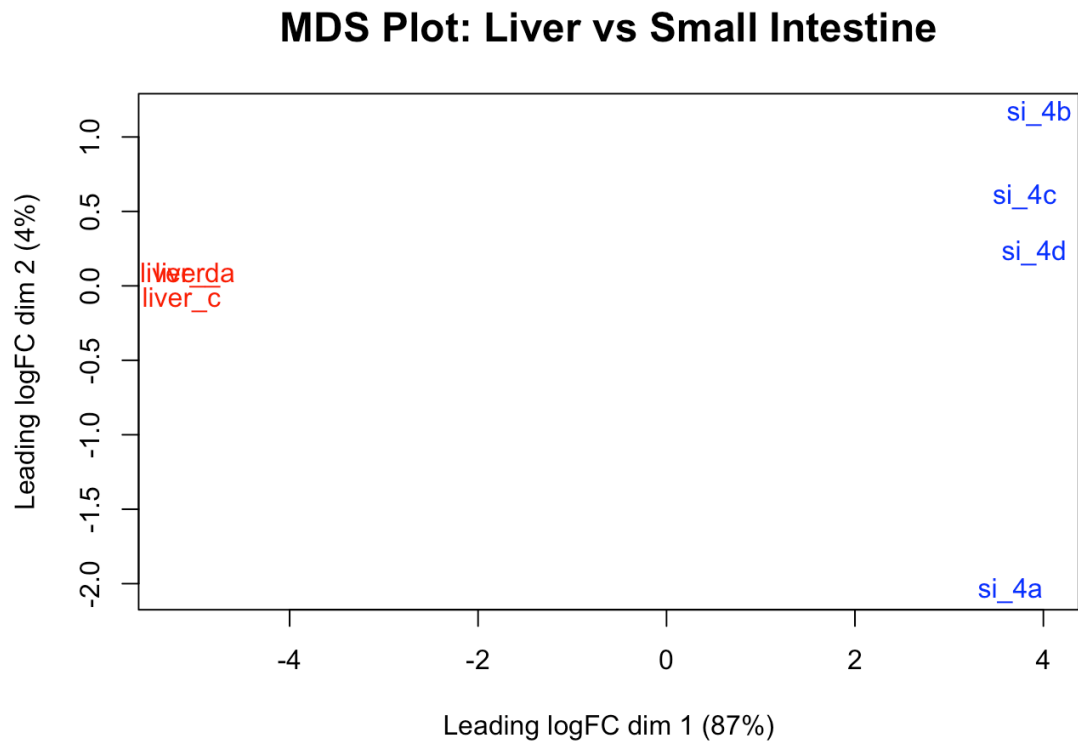


Figure 1. Multidimensional scaling (plot) of gene expression profiles for liver and small intestine tissue. Samples cluster by tissue type along the leading logFC dimension 1(87%). Dimension 2 shows minor group variability within small intestine samples.

Biological Coefficient of Variation (BCV) Plot

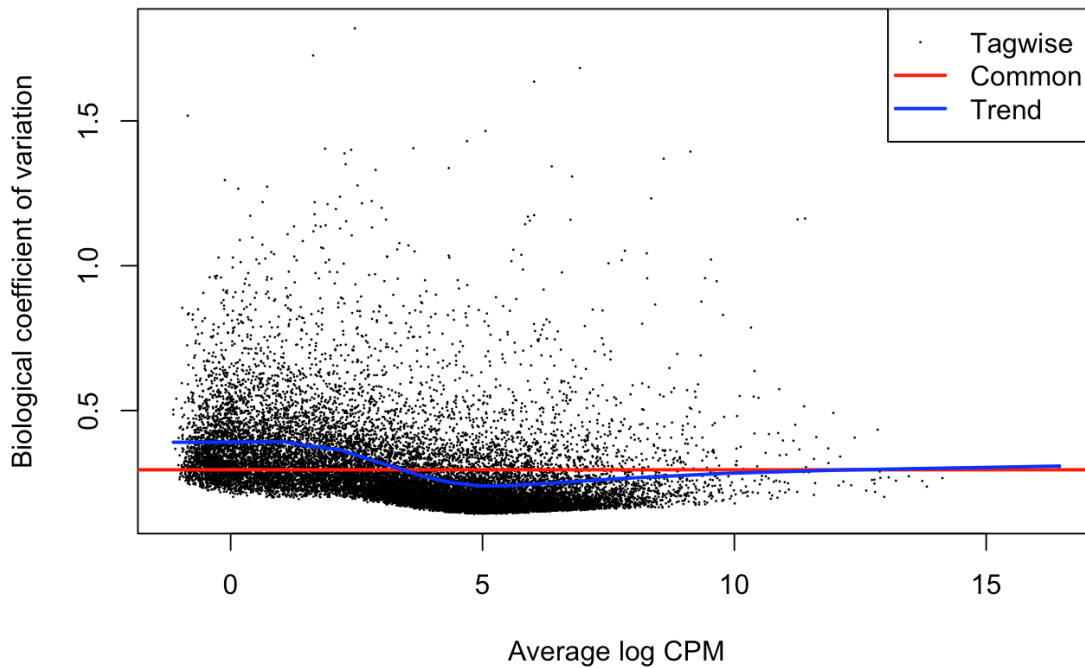


Figure 2. Biological coefficient of variation plot showing gene-wise dispersion (sample variance) relative to strength of average gene expression. Variability decreases as expression level increases, with higher dispersion seen in genes of lower expression.

Differential gene expression.

Differential expression analysis identified 11,300 significantly differentially expressed genes between the healthy liver tissue and healthy small intestine tissue ($FDR < 0.05$), 5,189 genes were upregulated in the liver and 6,111 genes were upregulated in the small intestine (Table 2). Smear plots from exact test (Fig.3.) and generalized linear models (GLM) (Fig.4) show a wide distribution of genes in both the negative and positive direction. Significantly differently expressed genes are highlighted in red. This indicates tissue-specific differentiation. Genes that

had positive log fold changes (LogFC) were more highly expressed in the small intestine, while genes with negative log FC were more highly expressed in the liver.

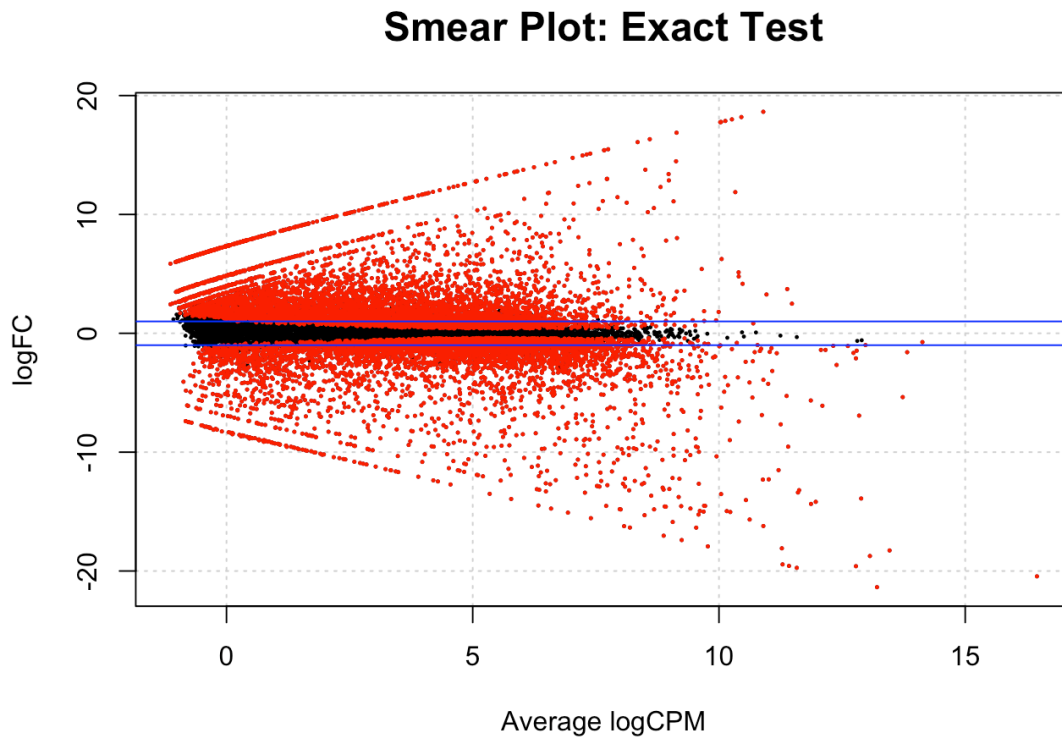


Figure 3. Smear plot of differential expression results from exact test comparing liver and small intestine samples. Each point represents a gene. Genes with significant differential expression as determined by $FDR < 0.05$ are highlighted in red.

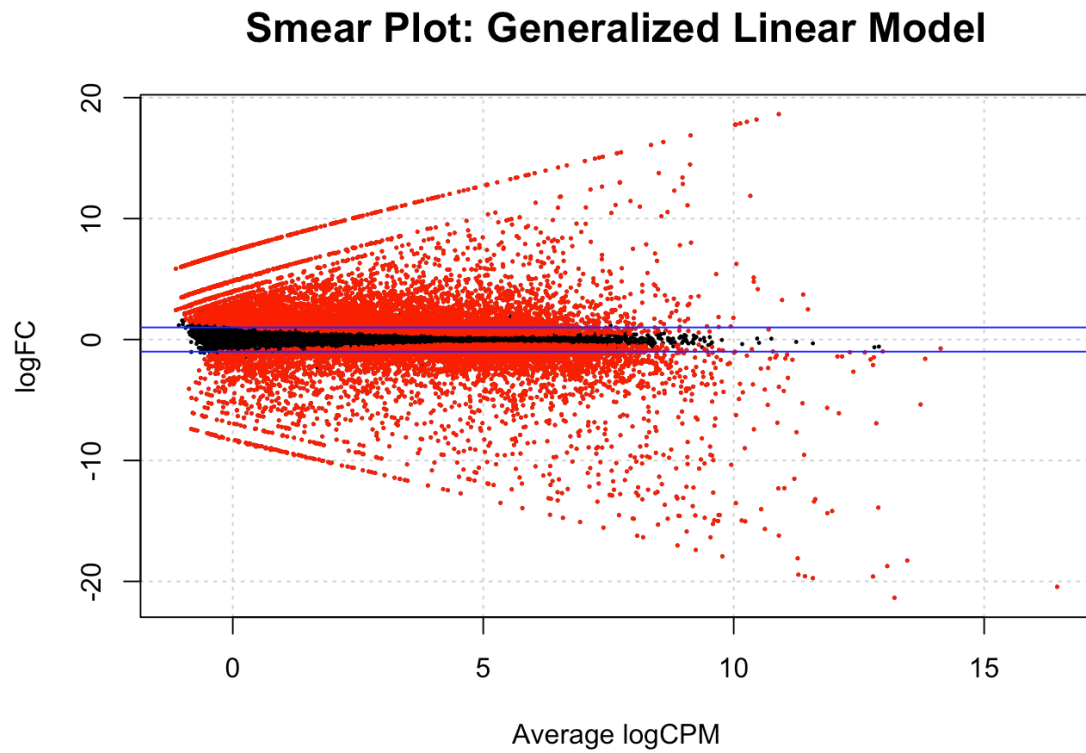


Figure 4. Smear plot Generalized Linear Model (GLM)analysis. Genes are represented by dots plotted by logFC vs average gene expression (logCPM). Genes that are significantly expresses (FDR<0.05) are highlighted in red.

Table 2. Summary of the significantly differentially expressed genes identified by edgeR (FDR<0.05).

	Number of genes
Upregulated in Small Intestine	6,111
Upregulated in Liver	5,189
Total Differentially Expressed	11,300

Highly Expressed Genes Tissue Specific Genes

Several genes showed significant tissue specific expression (Table 3). As expected, genes involved with digestion and immune defense were highly expressed in the small intestine. The top differentially expressed genes that were identified by GLM analysis were mainly from liver tissue, the strongest being FGA, C9, HPX, F9, FGG, C4BPA, F2, HP, ALB, and ANGPTL3. All of these genes are known to be involved with liver functions, such as oxidative stress (HP), transporting molecules (ALB), and immune response to foreign substances (C4BPA (regulates complement system)). Small intestine tissue has highly-expressed genes involved in digestion, nutrient transport and absorption (SI, FABP2) and mucous defense (AGR2) and defense against microorganisms (ITLN2, DEFA5, DEFA6) (Bevins and Salzman, 2011). Together, these results support that the transcriptome of small intestinal tissue is related to digestion and defense.

A heat map of the top 50 most differentially expressed genes (Fig. 5) clearly shows separation of small intestine from liver. The four small intestine tissue samples and the 3 liver samples clustered together. There was a clear differentiation between genes highly expressed in the liver and small intestine. Genes that were highly expressed in the liver were very lowly expressed in the small intestine and vice versa. This supports tissue-specific gene regulation and is consistent with the separation observed in the MDS plot. One sample from the liver (liver_a) showed a slight difference in expression from the other liver samples, particularly for the OLFM4 and CRP genes. Liver_a showed lower OLFM4 expression and higher CRP expression than the other liver samples.

Table 3a. Top Differentially Expressed Genes in Liver

	Gene	logFC	logCPM	LR	PValue	FDR
FGA	FGA	-21.36	13.21	1177.403	0	0
C9	C9	-16.35	9.54	1162.813	0	0
HPX	HPX	-12.29	11.01	1159.450	0	0
F9	F9	-17.03	8.88	1120.842	0	0
FGG	FGG	-19.59	12.78	1101.957	0	0
C4BPA	C4BPA	-14.51	9.70	1067.191	0	0
F2	F2	-15.01	9.69	1058.634	0	0
HP	HP	-18.27	13.46	1054.158	0	0
ALB	ALB	-20.45	16.46	1051.104	0	0
ANGPTL3	ANGPTL3	-14.57	8.90	1036.246	0	0

Table 3b. Top Differentially Expressed Genes in Small Intestine.

	Gene	logFC	logCPM	LR	PValue	FDR
REG3A	REG3A	18.00	10.26	909.6551	0	0
DEFA5	DEFA5	18.64	10.90	849.3463	0	0
SI	SI	17.76	10.03	845.9745	0	0
DEFA6	DEFA6	17.87	10.13	773.7954	0	0
KRT20	KRT20	16.87	9.14	730.4756	0	0
CHP2	CHP2	15.05	7.31	616.9464	0	0
EPCAM	EPCAM	7.80	8.93	558.6215	0	0
ITLN2	ITLN2	10.92	7.05	558.0147	0	0
FAM83E	FAM83E	8.40	5.71	541.8431	0	0
AGR2	AGR2	11.13	7.64	532.4323	0	0

Table 3. a. Top ten genes upregulated in the liver relative to small intestine tissue. Genes are ranked by FDR. negative logFC values indicated higher expression. Upregulated genes include some involved in coagulation and complement pathways, and transport, know functions of liver.

b. Top ten genes upregulated in the small intestine relative to liver tissue. The genes are ranked by false discovery rate and all have positive log fold change, indicating higher expression in

small intestine samples. Sucrase-isomaltase (SI), DEFA5, DEFA6, REG3A, ITLN2, and AGR2 are known to be associated with the small intestine.

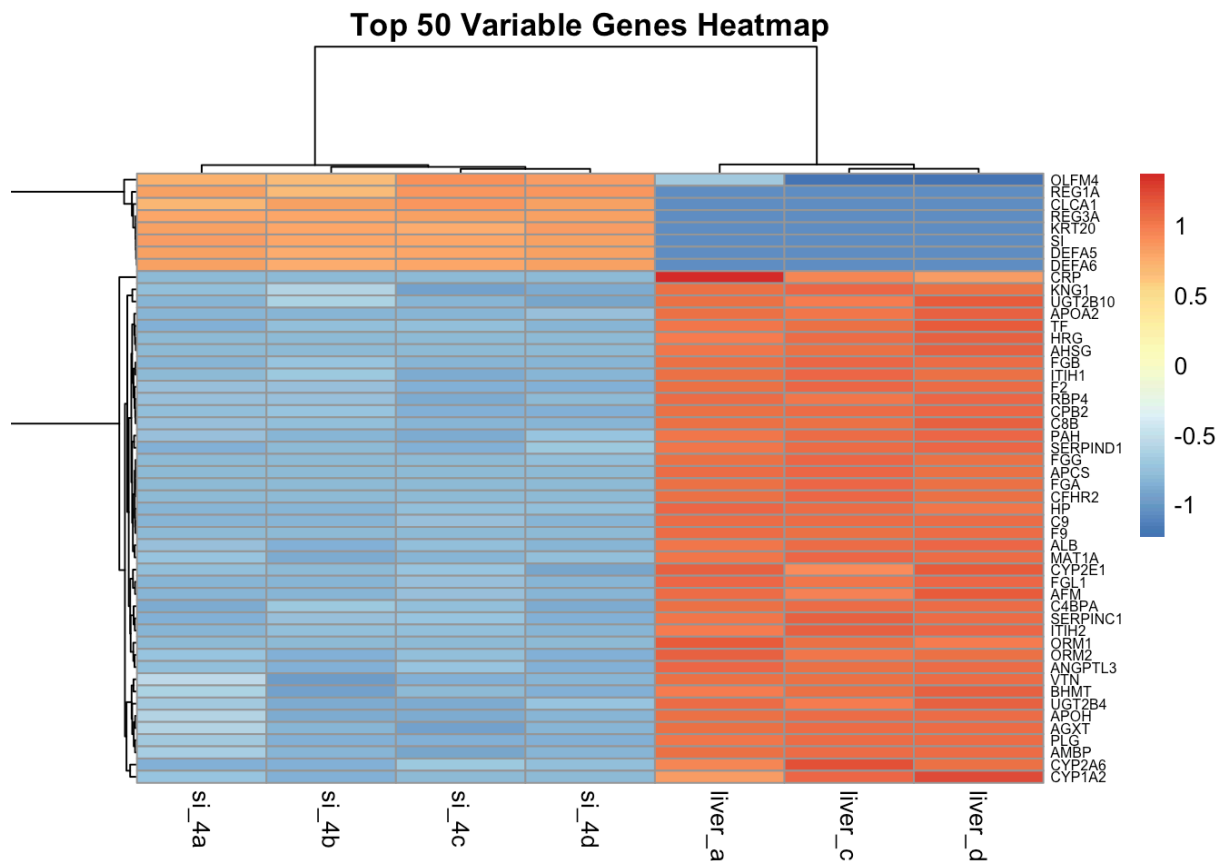


Figure 5. Heat map of the fifty most variable genes across liver and small intestine tissue samples. Gene expression values (logCPM) were scaled by gene and colored to highlight relative differences in expression (orange = high expression/blue = low expression). Samples cluster by tissue type and gene expression of tissue-specific genes is inversely related (the genes most highly differentially expressed in liver samples are lowest in small intestine). One sample

of liver tissue (liver_a) shows some divergence from the other replicates in some genes, most notably CRP and OLFM4.

Functional Enrichment Analysis

Functional enrichment analysis supported the biological differences between liver and small intestine gene expression. Genes that were highly expressed in healthy liver tissue were enriched for GO biological processes related to protein activation cascades (Fig. 6a), such as zymogen activation, plasminogen activation, fibrinolysis, and blood coagulation. KEGG pathways also identified cascades in coagulation and the complement cascade of the immune system (Fig. 6b). These processes and pathways support the liver's known role in inflammatory regulation and metabolic processing.

Genes upregulated in the small intestine were enriched for GO biological pathways associated with mucosal and immune defence and responses to bacterial infections (both gram negative and gram positive) (Fig. 7a). KEGG pathways associated with pathogen detection and response were enriched (Fig. 7b). Together these reflect the small intestines role as a barrier and in defense

against pathogens.

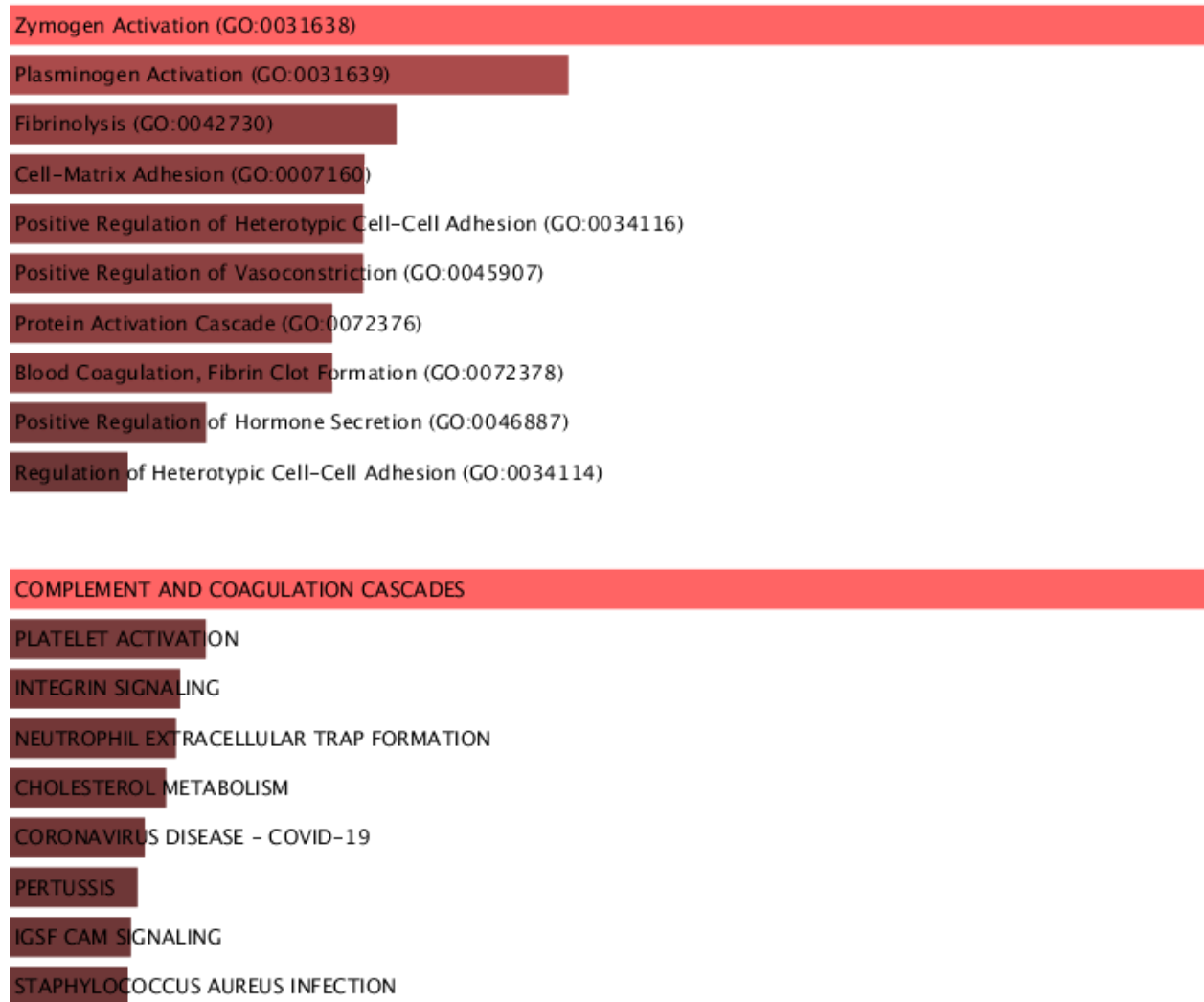


Figure 6. a. Gene Ontology Biological Processes enriched in liver tissue. b. KEGG pathways enriched in liver tissue.

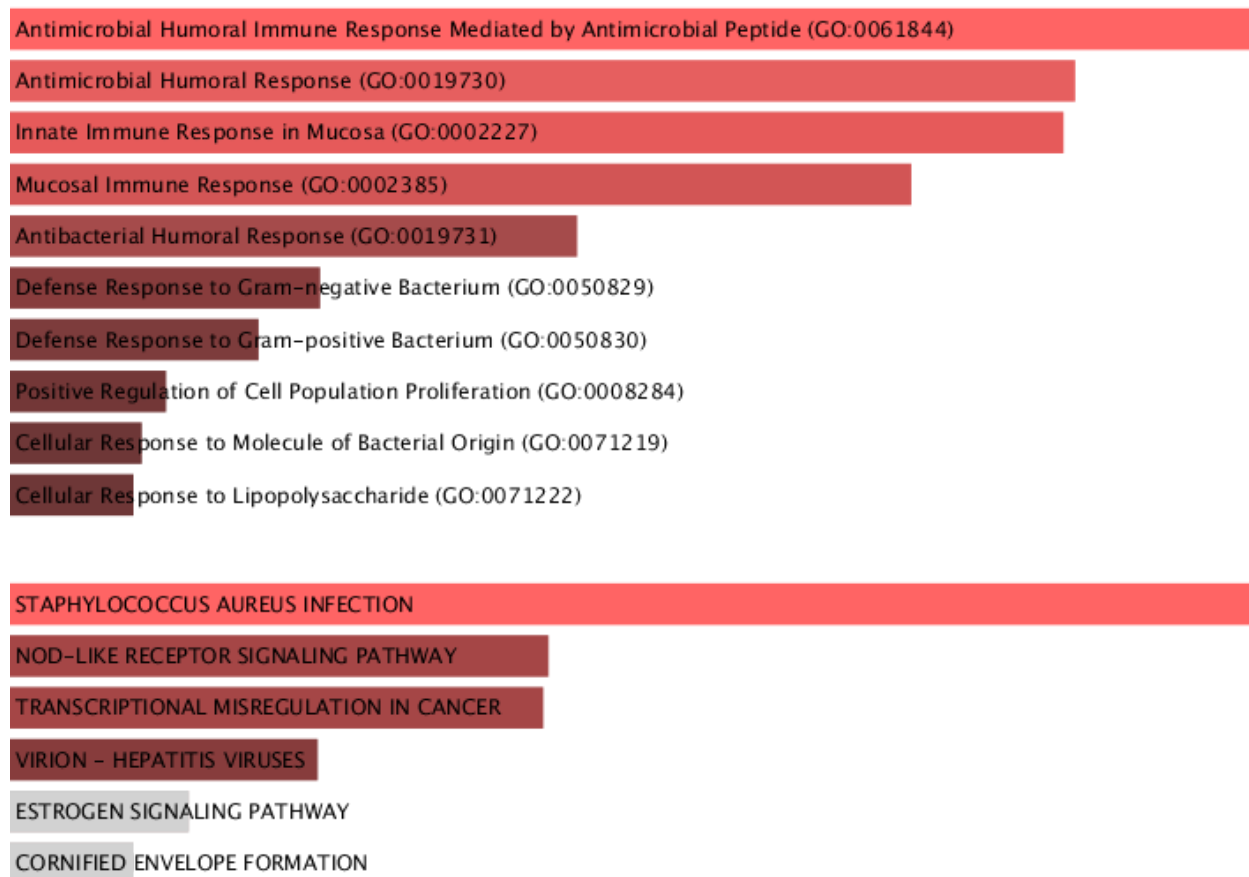


Figure 7. a. Gene Ontology Biological Processes enriched in healthy small intestine tissue. b. KEGG pathways enriched in healthy small intestine tissue.

These functional enrichment analysis results support strong tissue specialization due to specialized gene expression. The results provide evidence that the small intestine and liver may contribute different but complementary responses to microplastic exposure. The intestine being the primary site serving as a barrier, while the liver helps to process contaminants that enter the system.

Discussion

The goal of this project is to characterize baseline gene expression for two healthy tissues that would be important for responding to microplastic exposure. The small intestine was chosen because it is a primary site of exposure from potential ingested microplastics, and the liver was chosen because of its known role in metabolism of toxins and regulation of immune response.

Differential expression analysis identified a total of 11,300 significantly differentially expressed genes ($FDR < 0.05$) between the liver (5,189 higher in liver) and small intestine (6,111 higher in small intestine). These results, along with the clear clustering of samples in MDS and genes in the heatmap, support the distinct biological functions of each tissue type. Genes highly expressed in the small intestine included SI, FABP2, DEFA 5, DEFA6, and ITLN2. These genes have roles in digestion, barrier defence, pathogen immune response. Functional enrichment analysis provides further evidence that the pathways these proteins are involved in serve to help the small intestine function as a barrier and quickly respond to bacterial infection.

These findings are important considerations for microplastic exposure because ingested particles could enter the system in a similar manner. Furthermore, these pathways could be disrupted by microplastics and it would be important to have a baseline established before comparison to tissue post-exposure.

The liver transcriptome reflected its role in pathways that would be turned on in response to microplastic exposure after absorption through the small intestine. Upon reaching the liver, inflammatory cascades and metabolism may help to eliminate the contaminant. Previous reporting by Lee et al. (2014), has indicated that microplastic exposure changed fibrinogen levels and inflammatory gene expression, so it may have other physiological responses in addition to detoxification. One sample (liver_a) had a higher level of C-reactive protein compared to the

other liver replicates. CRP is produced by the liver in response to inflammation. This could indicate natural variation between individuals or it could indicate that the donor has some inflammation that was not diagnosed. Future studies with larger numbers of replicates could help to resolve whether this is a normal variation or if there is some physiological difference.

The transcriptome differences identified in this analysis support the model that the small intestine and liver would work together to respond to microplastic exposure. Initial response will occur in the small intestine, as it senses microplastics and first begins an immune response (along with its barrier function), and a subsequent response by the liver upon absorption into the blood stream. Establishing a baseline will provide a reference with which to make comparisons when exposed tissues are analysed.

This study only analysed healthy, unexposed tissue. Future studies should be done to compare these baseline data to tissues exposed to microplastics to identify exposure specific differentially expressed genes. This would allow for a better picture of systemic microplastic responses. Looking deeper into genes that have been implicated in microplastic exposure like those involved in oxidative stress and inflammation would help to get a better understanding of the exact mechanisms of tissue response or tissue damage from microplastic exposures. Investigation into the differences in gene expression due to variations in particle size (micro vs nano-particles) and composition would also be highly informative. In summary, this analysis provides a baseline transcriptome of healthy liver and small intestine tissues that provide an insight into their preparedness for microplastic exposure.

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